Linear Algebra I Midterm Solution

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1. (a) $\beta = \{(1,0,1,0,0),(0,1,0,0,0),(1,0,0,1,0),(0,0,0,0,1)\}$ is a basis for U. $\dim U = 4$.

   (b) $\gamma = \{(0,1,1,1,0),(1,0,0,0,-1)\}$ is a basis for W. $\dim W = 2$.

   (c) $U \cap W = \{(a_1,a_2,a_3,a_4,a_5) \in F^5 \mid a_2 = a_3 = a_4, a_1 = 2a_2, a_5 = -2a_2\}$. $\eta = \{(2,1,1,1,-2)\}$ is a basis for $U \cap W$. $\dim U + W = 1$. Therefore, $\dim U + W = 4 + 2 - 1 = 5 = \dim F^5$, $U + W = F^5$. The standard basis for F^5 is a basis for $U + W$.

2. (a) $T(1) = 1 + x + x^2, T(x) = 1, T(x^2) = 2x, T(x^3) = 3x^2$.

   (b) Suppose $f(x) = ax^3 + bx^2 + cx + d \in \ker T$, then

   $$3ax^2 + 2bx + c + d(1 + x + x^2) = 0.$$

   That is, $3a = 2b = c = -d$. A basis for $\ker T$ is $\{-\frac{1}{3}x^3 - \frac{1}{2}x^2 - x + 1\}$. $\text{nullity}\, T = 1$.

   (c) By the dimension formula, $\text{rank}\, T = \dim P_3(F) - \text{nullity}\, T = 4 - 1 = 3 = \dim P_2(T)$. Therefore, $\text{im}\, T = P_2(F)$. The standard basis for $P_2(F)$ is a basis for $\text{im}\, T$. T is not one-to-one since $\ker T \neq \{0\}$, T is onto since $\text{im}\, T = P_2(F)$.

3. (a) $T(1) = (1,1,1), T(x) = (-1,0,1), T(x^2) = (1,0,1)$, therefore

   $$[T]_\beta^\gamma = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

   (b) We claim that T is one-to-one, suppose $f(x) = ax^2 + bx + c \in \ker T$, then

   $$(a - b + c, c, a + b + c) = (0,0,0).$$

   This implies $a = b = c = 0$, i.e. f is the zero function, T is one-to-one. Since $\dim P_2(F) = \dim F^3 = 3$, and T is one-to-one, T is an isomorphism (thus invertible).

   (c) Suppose $g(x) = \alpha x^2 + \beta x + \gamma \in P_2(F)$ such that $T(g) = (a,b,c)$, then

   $$(\alpha - \beta + \gamma, \gamma, \alpha + \beta + \gamma) = (a,b,c).$$

   This implies $\alpha = \frac{a+c-2b}{2}, \beta = \frac{c-a-2b}{2}, \gamma = b$. Therefore,

   $$T^{-1}(a,b,c) = \frac{a + c - 2b}{2}x^2 + \frac{c - a}{2}x + b.$$

   is the inverse of T.

4. (a) $T((1,0)) = (1,3), T((0,1)) = (2,1), U((1,0)) = (1,0), U((0,1)) = (-1,2)$. Therefore,
$$[T]_\beta = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, [U]_\beta = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}.$$

(b) $[UT]_\beta = [U]_\beta[T]_\beta = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 6 & 2 \end{pmatrix}$. Similarly,
$[TU]_\beta = [T]_\beta[U]_\beta = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 3 & -1 \end{pmatrix}$

(c) We claim that the linear map is $L_{UT} : F^2 \to F^2, v \mapsto [UT]_\beta v$ is one-to-one, suppose $v = (x,y)^t \in \ker L_{UT}$, then
$$[UT]_\beta v = \begin{pmatrix} -2 & 1 \\ 6 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$
The solution to the equation can only be $(x,y) = (0,0)$, thus L_{UT} is one-to-one, and therefore invertible (the domain and the codomain of L_{UT} have the same finite dimension). Hence UT is invertible. The formula for $(UT)^{-1}$ is
$$(UT)^{-1}(x,y) = \left(\begin{pmatrix} -2 & 1 \\ 6 & 2 \end{pmatrix}^{-1}\begin{pmatrix} x \\ y \end{pmatrix}\right)^t = \left(\frac{-1}{5}x + \frac{1}{10}y, \frac{3}{5}x + \frac{1}{5}y\right).$$

5. (a) For any $x \in \ker T$, $T^2(x) = T(T(x)) = T(0) = 0$, so $\ker T \subseteq \ker T^2$. By assumption, $\operatorname{im} T^2 = \operatorname{im} T$. Thus, from the dimension formula
$$\dim V = \dim \operatorname{im} T + \dim \ker T = \dim \operatorname{im} T^2 + \dim \ker T^2.$$
$\ker T = \ker T^2$. If $x \in \operatorname{im} T \cap \ker T$, then $x = T(y)$ for some $y \in V$, and $T(x) = T^2(y) = 0$, thus $y \in \ker T^2 = \ker T$, i.e. $x = T(y) = 0$. Therefore, $\operatorname{im} T \cap \ker T = \{0\}$.
It is clear that $\operatorname{im} T + \ker T$ is a subspace of V, and
$$\dim(\operatorname{im} T + \ker T) = \dim \operatorname{im} T + \dim \ker T - \underbrace{\dim(\operatorname{im} T \cap \ker T)}_{0} = \dim V.$$
Therefore, $V = \operatorname{im} T \oplus \ker T$.

(b) Note that we have a filtration of subspaces
$$\cdots \subseteq \operatorname{im} T^n \subseteq \operatorname{im} T^{n-1} \subseteq \cdots \subseteq \operatorname{im} T^2 \subseteq \operatorname{im} T \subseteq V$$
Which means that we have a chain
$$\cdots \leq \operatorname{rank} T^n \leq \operatorname{rank} T^{n-1} \leq \cdots \leq \operatorname{rank} T^2 \leq \operatorname{rank} T \leq \dim V$$
Since $\dim V$ is finite, the sequence must terminate, i.e. there must be $k \in \mathbb{N}$ such that $\operatorname{rank} T^k = \operatorname{rank} T^{k+m}$ for any $m \geq 0$. In particular, $\operatorname{rank} T^k = \operatorname{rank} T^{2k}$, that is, $\operatorname{rank}(T^k)^2 = \operatorname{rank} T^k$, since T^k is linear, from 5-(a), we have
$$V = \operatorname{im} T^k + \ker T^k.$$

6. (a) Since V is finite dimensional, W is also finite dimensional. Let γ be a basis for W and extend γ to be a basis β for V. Put $\eta = \beta \setminus \gamma$, then define $U = \mathrm{span}(\eta)$. U is a subspace of V and $V = W_1 \oplus W_2$ since $\beta = \gamma \coprod \eta$.

   (b) Since $P^2 = P$, $\mathrm{rank}\, P^2 = \mathrm{rank}\, P$, then by 5-(a), $V = \mathrm{im}\, P \oplus \ker P$.